Uncertainty-calibrated graph models of multi-omics markers for immunotherapy response

Cambridge Computer Lab (Lio) extension - bridges the preprint line toward clinical AI · Gabriele Bambini · draft v1 · 2026-08-23

HYPOTHESIS

Response-to-checkpoint models fail clinically because point predictions hide uncertainty. Patient-graph ensembles with conformal risk control can output calibrated response SETS that clinicians can act on, while graph attribution exposes which omics modules drive each call.

AIMS

A1 patient-graph construction from TCGA+ICB labels | A2 ensemble+conformal model vs TIDE/TMB baselines | A3 calibrated response sets + attribution analysis -> workshop submission.

METHODS

Cohorts: TCGA multi-omics via GDC + public ICB response sets; features: TMB, expression signatures, pathway activity graphs. Models: deep ensembles inside a sheaf/GNN scaffold; conformal wrappers; baselines: TIDE-style signatures and TMB alone. Reporting: coverage curves, set sizes, attribution stability across seeds.

EXPECTED OUTPUTS

Workshop paper on calibrated graph models for ICB response; open evaluation harness; bridge to CCAIM application narrative.

BIBLIOGRAPHY & CHAIN OF VERIFICATION (2/8 SOURCES LIVE ON 2026-08-23)

Claim used	Source	Link	CoV
Pan-tumor genomic biomarkers correlate with PD-1 response	Cristescu et al., Science 2018	https://doi.org/10.1126/science.aar3593	◎ **unverified**
Tumour mutational load predicts survival under ICB	Samstein et al., Nat Genet 2019	https://doi.org/10.1038/s41588-018-0312-8	◎ **unverified**
TIDE-style expression signature predicts immunotherapy failure	Jiang et al., Nature Medicine 2018	https://doi.org/10.1038/s41591-018-0136-1	◎ **unverified**
Deep ensembles give practical predictive uncertainty	Lakshminarayanan et al., NeurIPS 2017	https://arxiv.org/abs/1612.01474	✓ **HTTP 200**

Conformal prediction wrappers for calibrated sets	Angelopoulos & Bates, 2021	https://arxiv.org/abs/2107.07511	**✓ HTTP 200**
TCGA multi-omics cohorts via GDC portal	NCI GDC	https://portal.gdc.cancer.gov/	**◎ unverified**
[CoV] GDC API status endpoint reachable	data source check	https://api.gdc.cancer.gov/status	**◎ unverified**
[CoV] TCGA programme page live	data source check	https://www.cancer.gov/about-nci/organization/ccg/research/structural-genomics/tcga	**◎ unverified**

Chain-of-verification method: every claim row was probed automatically at build time (2026-08-23); ✓ = endpoint answered HTTP 200 during generation, ◎ = probe failed or blocked (verify manually before citing in applications). Draft generated by the PhD Funding Command Center build.